

1 Physical Primitives of Mind: Modeling
2 Consciousness via Symmetries and Landauer
3 Erasure

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5 August 31, 2026

6 **Abstract**

7 We present a mathematical synthesis and formal Lean 4 verifi-
8 cation demonstrating how cognitive architectures can emerge as at-
9 tractor states from established physical primitives, without introduc-
10 ing new fundamental physics. Starting from the Poincaré group of
11 spacetime symmetries and finite phase space, we construct a frame-
12 work for predictive processing. Spontaneous symmetry breaking gen-
13 erates necessary topological boundaries, and finite phase space sub-
14 jects these boundaries to Landauer’s thermodynamic erasure. Agency
15 arises when systems evolve along paths of least action, minimizing en-
16 tropy production via internal structural resonance with external per-

17 turbations. Scaling this dynamic to macroscopic levels aligns with
18 emerging paradigms in neuroscience—specifically classical electrody-
19 namic fields (ephaptic coupling)—and Kuramoto phase synchroniza-
20 tion, culminating in a topological fixed point: a unified “Self.” We
21 state the claim narrowly, because the subject matter invites a wider
22 reading than the work supports: no new physics is derived here. Every
23 physical result the chain invokes is established—Picard–Lindelöf, Kib-
24 ble’s classification of topological defects, Landauer’s erasure bound,
25 Gibbs’ inequality, Barbalat’s lemma, Sakaguchi’s threshold, Banach’s
26 fixed point. What is new is that they are stated in one formal lan-
27 guage, their consistency mechanically checked, and the points where
28 the chain relies on stipulation rather than derivation located exactly.
29 The development declares no axioms and every structure carrying
30 physical content is inhabited by a witness; the synchronization step is
31 proved from a dynamics for initial data inside an explicit basin and
32 assumed outside it, and the identification of a global section of the
33 probability sheaf with the unity of experience is a stipulation, marked
34 as such throughout rather than absorbed into the derivation. The
35 framework is nonetheless falsifiable in one place that is its own and
36 no rival’s: recovery from sudden awakening should follow a delayed
37 sigmoid rather than an exponential, because the variable gating it is
38 astrocytic volume on a timescale of seconds to minutes rather than cor-
39 tical phase realignment on a timescale of milliseconds. As a corollary,
40 we show why rigid von Neumann architectures (e.g., GPUs) are ther-

41 modynamically disadvantaged or structurally constrained from reach-
42 ing these specific continuous thermodynamic attractors.

43 **1 Introduction**

44 The pursuit of a physical theory of consciousness typically involves search-
45 ing for the biological correlates of mind or treating subjective experience as
46 an abstract computational algorithm. This paper asks a more fundamental
47 question: If we assume only basic physical primitives—continuous symme-
48 tries (the Poincaré group), geometry, and thermodynamics—is the emergence
49 of a unified, perceiving entity a natural physical attractor?

50 We propose a framework where the structural properties of consciousness
51 are derived as physical attractor states. Establishing a deductive chain from
52 basic principles, we demonstrate how continuous symmetries break to form
53 boundaries, how those boundaries dissipate free energy as non-equilibrium
54 steady states, and how their evolution along paths of least action results
55 in physical resonance with their environment. This eventually scales into
56 the macroscopic electromagnetic unity proposed by emerging field theories
57 of mind. To avoid critiques of “math theatre,” we delineate between literal
58 physical mechanisms (e.g., thermodynamic erasure via Landauer’s principle,
59 boundaries from spontaneous symmetry breaking) and theoretical mappings,
60 utilizing Lean 4 formalization as our mathematical anchor. We adopt a per-
61 spective aligned with Russellian Monism: objective measurements of macro-

62 scopic electromagnetic fields constitute the third-person “outside,” while the
63 system’s internal physical deformation to reach structural resonance consti-
64 tutes the first-person intrinsic experience.

65 **What is new here, and what is not**

66 A framework paper about consciousness invites a particular suspicion, and it
67 is better answered at the start than at the end. **No new physics is derived**
68 **here.** Every physical result the chain invokes is established, and most of
69 it is old: Picard–Lindelöf for the well-posedness of the coupled dynamics
70 (Coddington & Levinson, 1955); Kibble’s classification of topological defects
71 by the homotopy of the vacuum manifold (Kibble, 1976); Landauer’s bound
72 on the heat cost of erasure (Landauer, 1961); Gibbs’ inequality for the relative
73 entropy (Cover & Thomas, 2006); Barbalat’s lemma for the asymptotics of a
74 dissipating trajectory (Barbalat, 1959); the threshold $K_c = 2D$ for the noisy
75 mean-field Kuramoto model (Sakaguchi, 1988); the bound of Still, Sivak,
76 Bell and Crooks relating dissipation to nonpredictive information (Still et
77 al., 2012); the identity of Kawai, Parrondo and Van den Broeck equating
78 dissipated work with the relative entropy between forward and time-reversed
79 ensembles (Kawai et al., 2007); and Banach’s fixed-point theorem (Banach,
80 1922). A physicist reading this paper will recognize all of it and would
81 establish the point in three paragraphs if we did not. Conceding it costs
82 nothing.

83 What is new is of a different kind, and it is a smaller claim than a theory

84 of consciousness usually makes. These results have been stated in one formal
85 language, their mutual consistency has been mechanically checked, and the
86 places where the chain relies on stipulation rather than derivation have been
87 located exactly. The value of a machine-checked chain is not that any link is
88 novel; it is that no link can quietly borrow from another, that a hypothesis
89 cannot be smuggled in as a definition, and that the seams — where physics
90 stops and interpretation begins — are fixed in the source rather than nego-
91 tiable in the prose. Section 2.1 reports what the check cost: three of five
92 original axioms each proved **False**, from one shared root cause, and none of
93 the three had been caught by reading.

94 The framework is not thereby unfalsifiable, and one of its predictions is
95 its own rather than a rival's. Section 11 argues that recovery from sudden
96 awakening should follow a *delayed sigmoid* rather than an exponential, be-
97 cause the variable gating recovery is astrocytic volume and cerebrospinal-fluid
98 clearance, on a timescale of seconds to minutes, while the cortical rhythms
99 whose realignment is usually taken to constitute the recovery move in mil-
100 liseconds. That is a claim about functional form; it is fittable against data of
101 a kind already collected; and a purely electrical account of the same recovery
102 predicts a different form.

103 2 Formalization in Lean 4

104 A core contribution of this work is the verification of our deductive chain
 105 using the Lean 4 interactive theorem prover. This approach builds on recent
 106 efforts to formalize consciousness and brain semantics, such as the ‘OnlyOne’
 107 system (Scherf, 2026) and sheaf-theoretic models of neural manifolds (Inoué,
 108 2026). It is necessary to explicitly distinguish which claims have been math-
 109 ematically derived as theorems within the proof assistant, and which remain
 110 irreducible physical postulates. Table 1 maps the primary claims of this
 111 paper to their current formalization status.

Table 1: Mapping of physical claims to Lean 4 formalization status. The development declares **no axioms**: `#print axioms` on any result above reports only `propext`, `Classical.choice` and `Quot.sound`. Earlier drafts carried five axioms, three of them inconsistent; all five became class fields, so a postulate is an obligation on each instance rather than a global assertion (Section 2.1), and two of the five have since been eliminated entirely (Derivation 5). “Theorem” still does not mean “physically established”: the postulates live in hypotheses and instance obligations, and every such class is inhabited by a witness in `Examples.lean`, so no row is vacuous.

Conceptual Claim	Lean 4 Implementation	Status
Pointwise Vacuum Minimization	<code>pointwise_vacuum_of_global_min</code> (Phase1), potential energy defined as an integral; holds a.e., and a double-well witness proves the a.e. cannot be strengthened	Theorem
Landauer’s Principle	Derived from statistical mechanics (Phase3); the axiom-free reversibility core becomes a heat bound via the field <code>StatisticalMechanics.heat_eq</code>	Theorem + instance postulate
Symmetry Breaking implies Boundaries	Derived from pointwise potential minimization (Phase1); witnessed on a double well with a $\mathbb{Z}_2$ symmetry whose minimisation hypothesis is proved, not assumed	Theorem
Inevitability of Boundaries (π_0)	The domain-wall case of the Kibble mechanism (Phase1 §3): on a connected substrate, a continuous field taking values in two different connected components of the vacuum manifold must leave the vacuum somewhere (<code>exists_notMem_of_connectedComponentIn</code>). Witnessed on the double well, whose vacuum manifold is computed to be exactly $\{-1, 1\}$ and proved disconnected (Examples §11.1); the field class is non-empty and the forced wall is located. Higher homotopy (π_1 vortices, π_2 monopoles) available and not taken	Theorem

Conceptual Claim	Lean 4 Implementation	Status
Nonpredictive information bounds dissipation	Still, Sivak, Bell and Crooks' bound is the field <code>PredictiveDissipation.still.bound</code> (<code>Phase3.PredictiveThermodynamics</code>). Everything downstream is derived: mutual information is defined from Mathlib's <code>klDiv</code> ; the data processing inequality for the chain $X_t \rightarrow S_t \rightarrow S_{t+1}$ proves the nonpredictive information non-negative (<code>predictiveInfo.le.mutualInfo</code>), from which $W_{\text{diss}} \geq 0$ follows rather than being assumed; a dissipation budget bounds wasted memory, and zero dissipation forces every retained bit to be predictive. Witnessed in Examples §18 by two systems over one two-bit law, one permitted to dissipate nothing and one forced to dissipate its whole memory, with a counterexample showing the no-back-action hypothesis is load-bearing. This replaces the earlier “gradient descent on σ drives $D_{KL} \rightarrow 0$ ”, which was an invalid inference	Theorem + instance postulate
Structural Resonance KL Bound	Gibbs' inequality proved; the bound is the field <code>StructuralResonance.kl.bound</code> (<code>Phase3.KLBound</code>). Retained: the postulate is consistent and yields $\sigma \geq 0$. No limit may be read off it	Theorem + instance postulate
Structural Resonance via Least Action	Gradient flow of σ in coupling space (Phase8), linked to <code>entropy_production_rate</code> ; finite substrates	Theorem
Phase-locking minimizes entropy production	For a <i>reciprocal</i> coupling kernel, the phase-locked field has the least entropy production of <i>any</i> field (<code>phase_locked_minimizes_entropy_of_symm</code> , Phase8 §2a). The mean-matching condition the earlier form assumed over the comparison class is derived, not imposed: the coupling term is anti-symmetric under exchanging its two sites, so total drift is conserved across phase fields (<code>total_drift_eq_of_symm</code>). Reciprocity is necessary — a one-way coupling on two sites strictly beats its own phase-locked state (Examples §16)	Theorem
Electrodynamic Discretization	Lebesgue-Bochner edge weights (Phase2) sitting on faces of the complex and partitioning the region's stress-energy; mesh refinement converges, witnessed on $[0, 1)$. The scalar read off the stress-energy tensor is a marked modelling choice: the partition holds for any non-negative choice, and two instances differing only in it have different couplings (Examples §6)	Theorem + instance postulate

Conceptual Claim	Lean 4 Implementation	Status
Phase synchronization to Unity	Sheaf-theoretic gluing (Phase5) over a ThermodynamicCover , whose fields are now all local: the global section is constructed by probability_glue_unique and the former class field sync_to_section_eq is a theorem about it. Witnessed by two patches built from different mass profiles, each section constructed over its own patch, whose glued state is neither of them (Examples §14). The cover's equilibrium hypothesis is no longer assumed on every instance: ThermodynamicCover.ofConvergentTrajectory (Phase5.EquilibriumBridge) discharges it from kuramoto_tendsto_global_minimum , and Examples §17.1 is a three-patch cover whose phase field is the limit of a trajectory started <i>unsynchronized</i> . The overlap-agreement hypothesis is untouched	Theorem + instance postulate
Reflexive Topology (the Self)	Banach fixed point for a contracting self-prediction map (Phase6); metric, completeness and contraction are hypotheses, all three discharged on the three-site substrate (Examples §10), where global sections are proved to be exactly the finite measures on it, the metric is the uniform distance between their densities, and the self-prediction map contracts by exactly 1/2 without being constant. The avatar's link to the field is the predicate IsRestriction-Resonance , discharged on the same witness; under it a covering family of avatars determines the global state (eq_of_avatar_eq), and without it the Self theorem is unchanged (self_of_const-Resonance)	Theorem (witnessed)
Kuramoto Lyapunov descent	Descent chained to the minimum characterization by the rotating frame (Phase4.RotatingFrame); identical frequencies	Theorem

Conceptual Claim	Lean 4 Implementation	Status
Kuramoto dynamics run	The initial value problem is well posed on all of $\mathbb{R}$: the field is globally Lipschitz and globally bounded (<code>kuramotoField.lipschitz</code> , <code>kuramotoField.norm_le</code>), and through every state at every instant passes exactly one trajectory, defined for all time (<code>kuramoto_cauchy_problem</code>). Along <i>every</i> trajectory of a zero-frequency system the dynamic potential converges (<code>dynamic.potential.tendsto</code>) and the total dissipation $\int_0^\infty \sum_i \dot{\theta}_i^2$ is finite, equal to the potential's total drop (<code>dissipation.integral.tendsto</code>). The motion stops: Barbalat's lemma is proved here, Mathlib lacking it, and applied, so $\sum_i \dot{\theta}_i^2 \rightarrow 0$ along every trajectory (<code>velocity_sq.tendsto.zero</code>). Where it stops is proved under an initial-data condition: if the phases start within a quarter turn and the potential excess starts below $a/2$, where $a > 0$ bounds the coupling below, the trajectory converges to a phase-locked configuration that is a <i>global</i> minimiser, with $r^2 = 1$ (<code>kuramoto.tendsto.global.minimum</code>), by a Polyak–Łojasiewicz estimate rather than a LaSalle principle. Unconditionally this is false — splay and twisted states are equilibria — so the condition cannot be removed. Witnessed on three sites, where no closed form exists (<code>Examples §17</code>)	Theorem (conditional)
Continuous vs. Rigid Topology	Fixed wiring support at matched resource (<code>Phase7.Rigidity</code>), deriving the suboptimality Phase 7 assumed; typed continuum/discrete separation	Theorem
Critical Coupling Threshold ($K_c = 2D$)	Self-consistency equation (<code>Phase8.SelfConsistency</code>): at or below K_c , threshold included, only $r = 0$ solves it; above K_c <i>exactly one</i> $r > 0$ does (<code>supercritical.fixed.point.existsUnique</code>), so the non-negative solutions are $\{0\}$ below and $\{0, r\}$ above; that r tends to 0 as $K \downarrow K_c$ and increases strictly with K (<code>coherent.branch.strictMono</code>); a solution is a density reproducing its own order parameter. Conditional on the von Mises stationary density, which is an input, and on no link to a trajectory	Theorem (conditional)

112 2.1 Soundness of the formalization

113 Formalization is only as trustworthy as the axioms it rests on, and an axiom
114 that is inconsistent silently makes every theorem downstream of it vacuous

115 while leaving the build green. We report a case of exactly this, found by
 116 auditing our own development, because the failure mode is easy to reproduce
 117 and we have not seen it discussed in the applied-formalization literature.

118 Earlier drafts of this work declared five axioms. Three of them were
 119 individually inconsistent: each, on its own, proves **False**. All three shared
 120 one root cause — an axiom that constrains a symbol it does not itself bind.
 121 Concretely:

- 122 • `spontaneous_symmetry_breaking_pointwise_min` took the potential-
 123 energy functional as an *implicit* argument occurring only in its hypoth-
 124 esis `PotentialEnergy phi = V v0`. Instantiating that functional with
 125 the constant map $\lambda_. V(v_0)$ discharges the hypothesis by `rfl` and yields
 126 the conclusion for an arbitrary field. With $V(v) = v^2$, $v_0 = 0$ and $\phi \equiv 1$
 127 this proves $1 = 0$.

- 128 • `landauer_heat_eq` equated the free class field `heat_dissipation` with
 129 a bath-entropy difference *for every instance* of the class. An instance
 130 with zero heat dissipation and a bath that gains one bit gives $0 = \log 2$.

- 131 • `kl_bound_axiom` asserted $\sigma \geq D_{KL}(P||Q)/\Delta t$ for *all* P and Q , while σ
 132 is a single real fixed by a class field. Since D_{KL} is unbounded above, a
 133 point mass against a uniform distribution refutes it.

134 The remaining two axioms, `phase_invariant_periodic` and `sync_to_section_eq`,
 135 had the same shape — global assertions pinning free fields of `LocalSectionSynchronization`

— and are refutable as soon as the probability presheaf admits two distinct global sections. (Both were demoted to class fields at the time; both have since been removed from the class altogether, for a separate reason given in Derivation 5.)

The correction is uniform, and we state it as a rule we now follow: *a physical postulate that mentions a class field must be a field of that class, never a standalone axiom quantified over all instances.* A postulate should be an obligation each model discharges, not a global claim about every model. Recast this way, the development declares no axioms at all. Two of the five became genuine theorems — pointwise vacuum minimization is now proved from $\int f = 0 \wedge f \geq 0 \Rightarrow f = 0$ a.e., and Gibbs’ inequality was already one — and the other three became instance obligations.

This makes non-vacuity the live question, since an uninhabitable class makes its theorems vacuous just as surely as an inconsistent axiom did. **Examples.lean** therefore constructs witnesses: a one-bit erasure system with a genuine bath that discharges **heat_eq** and reproduces $\Delta Q = k_B T \ln 2$; a perfectly resonant system discharging **kl_bound**; and a scalar field on a one-point spacetime discharging **ActionPrinciples**. Derivation 5’s gluing structures are witnessed too, which was until recently the largest open gap: **Examples.lean** builds a **ThermodynamicCover** over a three-site substrate carrying the discrete topology and its Borel σ -algebra, covered by two *overlapping* open patches whose intersection is a single shared site. The local sections are germ families of a phase-indexed finite measure $\mu_\theta =$

159 $(1 + \cos \theta)^+ \delta_{\text{mid}}$ pushed into the sheafification of the probability presheaf; the
 160 family is provably non-constant in θ (`phaseMeasure_not_const`), so the pe-
 161 riodicity obligation of the constructor `LocalSectionSynchronization.of-`
 162 `InvariantMeasure` is discharged by an argument about `cos` rather than satis-
 163 fied trivially, and `thermodynamic_equilibrium` is *derived* from `phase_locked_minimizes_potential`
 164 rather than assumed. `global_section_from_thermodynamics` then fires on
 165 this instance and glues the two patch-local sections into a unique global one.
 166 Two further covers answer the follow-up question of how non-degenerate such
 167 a witness can be made. `Examples.lean` §13 has a negative half worth record-
 168 ing: the phase field can be made non-constant as a function, but not as a
 169 physical state, because `thermodynamic_equilibrium` forces every instance
 170 to a phase-locked configuration. `Examples.lean` §14 is built the other way
 171 round, from two patches with independently chosen mass profiles that meet
 172 only on the site they share, and the section the gluing returns is neither of
 173 them. That is the same kind of result as `contracting_implies_const` in
 174 Derivation 6 — a proof that the witness is as strong as the class permits,
 175 rather than a witness that could be improved by trying harder.

176 What the witnesses establish is that the classes are inhabitable, so the
 177 theorems about them are not vacuous. What they do not establish is that
 178 a cortex is such an instance. In the cover above the phase field is already
 179 constant — every patch locked at $\theta = 0$ — which is precisely what makes
 180 the equilibrium obligation dischargeable; getting a physical system into that
 181 configuration is the content of the informal dynamical argument, not of the

182 Lean development.

183 The same audit turned up a defect of a different kind, worth record-
184 ing because it is not an axiom problem and would not have been caught
185 by `#print axioms. mesh_refinement_convergence` — the claim that dis-
186 crete edge weights converge to the continuous energy as the mesh is refined
187 — sat in a “Conjectures” section as an unproved `def ...: Prop`. It was
188 not merely unproved but false. Nothing required the edge regions to cover
189 the manifold, and $\text{diam } \emptyset = 0$, so the triangulation whose edge regions are
190 all empty satisfied the mesh hypothesis while contributing nothing to the
191 sum, forcing $|0 - \int f| < \varepsilon$ for every $\varepsilon > 0$. Its binder was dead as well: it
192 quantified over a triangulation and then resolved the vertex type by instance
193 search rather than through the bound variable, so the variable was never
194 used. `Phase2_MeshConvergence.lean` replaces it with a theorem. Conver-
195 gence is now stated along a *sequence* of triangulations of a named region
196 S , each required to be regular — edge regions measurable, without self-
197 loops, disjoint across distinct edges, covering S , and anchored to the sim-
198 plicial data in the sense that a vertex lies in the closure of every region
199 of an edge at it. Closure rather than membership is forced: cells are dis-
200 joint, so a shared vertex belongs to at most one of them. The manuscript’s
201 discrete energy $\frac{1}{2} \sum_u \sum_v w(u, v) f(u)$, which samples only the first endpoint
202 and is therefore not symmetric, is shown to equal exactly the midpoint rule
203 $\sum_{u < v} w(u, v) \frac{1}{2} (f(u) + f(v))$ over unordered edges; without that identifica-
204 tion, reading the double sum as a Riemann sum would be a category error.

205 At fixed mesh the error is at most $\varepsilon \mu(S)$ with ε a modulus of continuity of
 206 the integrand, and convergence follows for uniformly continuous integrands
 207 as the mesh shrinks. The regularity conditions pull against each other, so
 208 `Examples.lean` §6 discharges them with the uniform partition of $[0, 1)$ into
 209 half-open cells and checks that the approximation genuinely moves $-\frac{1}{2}$ at
 210 $N = 1$ and $\frac{1}{4}$ at $N = 2$ for the tent function $x \mapsto |x - \frac{1}{2}|$ — rather than
 211 converging trivially. What is proved is convergence, not the $O(1/N^2)$ rate
 212 the accompanying simulation measures; that would need a midpoint error
 213 term with a second derivative, which we have not developed.

214 A third defect was one of bookkeeping across files, and it is the kind that
 215 survives any number of green builds because each half is individually cor-
 216 rect. The development carried *two* Kuramoto potentials. Lyapunov descent
 217 along the dynamics, $\dot{V} = -\sum_i \dot{\theta}_i^2$, was proved for the full potential $V(\theta) =$
 218 $-\frac{1}{2} \sum_{i,j} A_{ij} \cos(\theta_i - \theta_j) - \sum_i \omega_i \theta_i$, while the characterization of the minimum
 219 — phase-locked in both directions — and everything downstream of it, in-
 220 cluding Derivation 5’s gluing, concerned the coupling term alone. Nothing
 221 chained them, and the gap was not cosmetic: `kuramoto_potential_unbounded_below`
 222 now proves that as soon as one $\omega_i \neq 0$ the full potential is unbounded below
 223 — walk θ out along ω and the bounded cosine sum cannot compensate — so
 224 it has no minimum for phase-locking to attain, and the sentence “the phase-
 225 locked state minimizes the Lyapunov potential” was, read literally, false of
 226 the functional the descent theorem was about. `Phase4_RotatingFrame.lean`
 227 closes this by formalizing the standard reduction $\theta_i \mapsto \theta_i - \Omega t$. For identical

228 natural frequencies $\omega \equiv \Omega$, a trajectory maps to a trajectory of the zero-
 229 frequency system — which is exactly the system Derivation 5 already worked
 230 with — the two potentials coincide there, and the descent result transports
 231 onto the coupling potential whose minimizers are the phase-locked states.
 232 Phase differences, phase-locking and the magnitude of the order parameter
 233 are all proved invariant under the change of frame, so nothing observable
 234 is lost in the reduction. `Examples.lean` §7 runs the chain end to end on
 235 two oscillators with a common frequency, discharging every hypothesis. The
 236 restriction is real and we state it plainly: the reduction is exact only for
 237 identical frequencies. A genuine spread of natural frequencies leaves residual
 238 detunings in the reduced system, phase-locking then requires coupling above
 239 a critical K_c , and K_c is nowhere in our Lean development — it is studied
 240 numerically, as Table 1 records.

241 A fourth defect was dimensional, and it concerned the one row of Table 1
 242 that is still numerical rather than proved. `Phase8_ContinuousField.lean`
 243 defines the critical coupling $K_c = 2D$ and a predicate for being above it.
 244 That predicate compared $\int_x \int_y K(x, y)$ against $2D$ — but a double integral
 245 against an unnormalized measure carries the substrate’s mass twice over,
 246 so a constant kernel K on a substrate of total mass m contributed Km^2 .
 247 The comparison therefore depended on how large the substrate was rather
 248 than on how strongly it was coupled, and any kernel whatever could be
 249 pushed above threshold by inflating m . `Examples.lean` §9 exhibits the fail-
 250 ure concretely: on a two-site substrate carrying counting measure, a sys-

251 tem coupled at $K = 1/2$ against a threshold of $K_c = 1$ — a factor of two
 252 *below* threshold — had an unnormalized double integral of $2 > 1$ and so
 253 satisfied the old predicate. The fix is to require the substrate’s measure to
 254 be a probability measure, which is the normalization in which the mean-
 255 field derivation of $K_c = 2D$ is carried out; `mean_field_coupling_const`
 256 then gives that a constant kernel has mean-field strength exactly K , and
 257 `exhibits_phase_transition_const_iff` reduces the predicate to $K > 2D$.
 258 We flag what this does and does not do. It makes the predicate say something
 259 whose truth depends only on the physics, and §9 witnesses it on both sides of
 260 the threshold. It does not by itself prove $K_c = 2D$: the content of that value
 261 is the bifurcation of the self-consistency equation $r = I_1(Kr/D)/I_0(Kr/D)$
 262 at $K = 2D$, which is a separate question from whether the predicate is di-
 263 mensionally sound. That bifurcation is proved below, in both directions,
 264 from an assumed stationary density; deriving the density itself needs the
 265 Fokker–Planck operator and is not done.

266 We draw no grand moral from this beyond the obvious one. The value
 267 of a proof assistant here was not that it certified our argument, but that it
 268 made a specific, checkable claim about what had been assumed — and that
 269 claim turned out to be false in a way no amount of reading the prose would
 270 have revealed.

271 3 Axiom 1: Symmetries and Phase Space

272 Our starting premise relies on fundamental geometry. In our Lean formaliza-
273 tion (`Phase1Primitives.lean`), continuous spacetime symmetries (e.g., the
274 Poincaré group) act upon field states, and the invariance of the system’s total
275 energy under that action is recorded by the class `SymmetryInvariantAction`.
276 Physically, it is such an invariance that Noether’s theorem converts into the
277 conservation of energy and momentum. The scope of the *formal* claim is nar-
278 rower and we state it plainly: the invariance is a class field, i.e. an assumption
279 a model must discharge, and no conserved quantity is ever constructed from
280 it. There is no Noether theorem in our Lean, and nothing downstream of this
281 section depends on one. The class is now inhabited — `Examples.lean` §11
282 exhibits a double-well potential $V(v) = (v^2 - 1)^2$ on a two-point spacetime
283 with the $\mathbb{Z}_2$ action $v \mapsto \pm v$, discharges the invariance because $(\pm v)^2 = v^2$,
284 and proves the vacuum manifold $\{-1, 1\}$ degenerate with the two minima ex-
285 changed by the symmetry, so an energy-minimising field and its image have
286 equal energy and differ. That is symmetry breaking exhibited rather than
287 assumed; it is still not a conserved quantity, and $\mathbb{Z}_2$ being discrete there is
288 no one-parameter family to differentiate along in the first place. Simultane-
289 ously, discrete permutation symmetries dictate the statistical mechanics of
290 any physical system, defining a strictly finite volume of phase space.¹

¹While the Poincaré group and discrete permutations serve as accessible physical ex-
amples, the underlying Lean formalization does not hardcode specific physics. It defines
the system at a more primitive mathematical level, relying entirely on generic Pseudo-
Riemannian manifolds (for spacetime) and finite measure theory (for statistical mechan-

291 Because phase space is finite, any localized physical system possesses a
 292 mathematical limit on the amount of information (distinguishable states) it
 293 can embody at any given moment.

294 4 Derivation 1: Symmetry Breaking and the 295 Inevitability of Boundaries

296 A perfectly symmetric universe is the uniform vacuum; no distinct entities
 297 exist. For any dynamic process to occur, the continuous symmetries of the
 298 background field must undergo spontaneous symmetry breaking.

299 When a field drops to a lower, asymmetric energy state, topology dic-
 300 tates the natural creation of defects (such as domain walls or monopoles).
 301 This constraint necessitates the creation of a physical boundary—a localized
 302 topological distinction between an “inside” and an “outside.” While such
 303 a defect (e.g., a crack in a crystal lattice) is necessary to define a separate
 304 entity, it is not sufficient for agency. The boundary is relational, defined by
 305 the twisted geometry of the field surrounding it, and must be coupled with
 306 non-equilibrium thermodynamics to exhibit active behaviors.

307 That first sentence was, until recently, the one claim of this section that

ics). What the formalization supplies is the vocabulary in which symmetries, conserved quantities and phase-space constraints can be *stated*; it does not derive them. All three of the classes involved are now inhabited in `Examples.lean` — the $\mathbb{Z}_2$ action and its invariance in §11, the real line with the Euclidean metric in §12 — but the instances are witnesses of inhabitability, not of physics: no theorem in the development consumes a symmetry group or a metric, so this section still contributes vocabulary rather than content.

308 the formalization did not make. What Lean established was the energy half
 309 — an energy-minimizing field sits in the vacuum manifold almost everywhere
 310 (`pointwise_vacuum_of_global_min`) — together with the observation that
 311 a boundary defect *obstructs* extension over a contractible interior. Neither
 312 produces a defect, so “topology dictates the natural creation” was asserted
 313 rather than proved. It is now the theorem `exists_notMem_of_connected-`
 314 `ComponentIn`: if the substrate is connected and a continuous field takes val-
 315 ues in two *different connected components* of the vacuum manifold M , then
 316 somewhere the field is not in M at all. That excluded region is the wall.
 317 This is the π_0 case of Kibble’s classification of topological defects by the ho-
 318 motopy of the vacuum manifold (Kibble, 1976), and the proof is elementary
 319 in exactly the way it is elementary in the physics literature: the continuous
 320 image of a connected space is connected, a connected subset of M lies in-
 321 side one component of M , and the field’s range would be such a subset if it
 322 never left M . No hypothesis mentions energy, a potential, or a symmetry;
 323 connectedness of the substrate and disconnectedness of the vacuum are the
 324 whole of it.

325 We formalize the π_0 case and stop there. The higher cases — $\pi_1 \neq 0$
 326 forcing vortex lines, $\pi_2 \neq 0$ forcing monopoles — are the rest of Kibble’s
 327 classification and Mathlib’s homotopy-group library would support at least
 328 the next one; they are available and not taken, because the argument this
 329 paper runs uses domain walls and nothing else. The witness is the double
 330 well of `Examples.lean` §11.1, where the vacuum manifold is *computed* to be

331 exactly $\{-1, 1\}$ (`wellVacuum_eq`) rather than merely known to contain both
 332 minima, and its disconnection is proved in the form the theorem consumes:
 333 no preconnected subset of it contains both, since a preconnected subset of $\mathbb{R}$
 334 is an interval and an interval spanning -1 and 1 contains 0 , which is the top
 335 of the barrier. `wellV_domain_wall` then quantifies over *every* continuous field
 336 on *any* connected substrate that reaches both minima, with no formula for
 337 the field anywhere in the statement. That the class is non-empty is checked
 338 separately, because a theorem quantified over all fields is worthless if none
 339 satisfies its hypotheses: the clipped identity on $\mathbb{R}$ does, and the wall it is
 340 forced to have sits at the origin (`kink_zero_notMem`).

341 **5 Derivation 2: The Thermodynamic Con-** 342 **straint**

343 This newly formed topological boundary is subjected to gauge perturbations
 344 from the “outside” environment. Because the internal phase space is finite
 345 (Axiom 1), it cannot absorb an infinite sequence of incoming perturbations
 346 and must overwrite old internal states.

347 By Landauer’s Principle (Landauer, 1961), the erasure of physical in-
 348 formation irreversibly decreases internal entropy, mandating that the sys-
 349 tem exhaust heat into the environment. Philosophers of physics (such as
 350 John Norton (Norton, 2005) and Orly Shenker (Shenker, 2000)) have cri-
 351 tiqued Landauer’s principle, arguing that physicists conflate *logical* erasure

352 (a many-to-one mapping in abstract computational states) with *thermody-*
353 *namic* erasure (a many-to-one mapping in physical phase space). They argue
354 that logical erasure could conceptually be implemented by an injective phys-
355 ical process without dissipating heat.

356 Our Lean 4 formalization (`Phase3.CombinatorialThermodynamics.lean`)
357 addresses this critique by defining erasure as a non-injective mapping on the
358 *physical* state space ($\neg \text{Function.Injective } t$). We formalize the global evolu-
359 tion of the bipartite universe (System + Bath) as strictly injective, repre-
360 senting Liouville’s theorem. Lean verifies that heat dissipation is not tied
361 to abstract logical erasure, but is a mandatory mathematical consequence of
362 any local physical state space contraction under globally reversible dynamics.
363 If a cognitive system’s internal topology forces a contraction of its physical
364 state space to accommodate structural resonance, heat dissipation becomes a
365 thermodynamic requirement. A stable boundary thus constitutes a dissipa-
366 tive structure (Prigogine & Stengers, 1984), operating as a non-equilibrium
367 steady state that consumes free energy to pay the thermodynamic cost of
368 erasure. Continuous thermal fluctuations play a constructive role in estab-
369 lishing boundaries between states while contributing to the system’s flexible,
370 adaptive response.

371 6 Derivation 3: Prediction as the Thermody- 372 namic Cost of Memory

373 A dissipative structure is driven by its environment. Write X_t for the state
374 of the structure and S_t for the environmental signal driving it, and let the
375 signal evolve by a dynamics of its own, $S_t \rightarrow S_{t+1}$, on which the structure
376 does not act. Two quantities are then defined at each step: the *memory*
377 $I(X_t; S_t)$, the mutual information between the structure's present state and
378 the signal's present value, and the *predictive information* $I(X_t; S_{t+1})$, which
379 is what that same state says about the signal's next value.

380 **What this section used to claim, and why it is withdrawn.** Earlier
381 versions of this derivation argued as follows. Model the environment as a
382 stochastic driving force $P(ext)$ and the structure's internal transitions as
383 $Q(int)$; entropy production σ is then bounded below by $D_{KL}(P||Q)/\Delta t$;
384 the principle of least action drives σ downward; therefore $D_{KL}(P||Q) \rightarrow$
385 0, and the internal transition probabilities come to mirror the statistical
386 symmetries of the environment. The final step does not follow. Rearranged,
387 the bound reads $D_{KL}(P||Q) \leq \Delta t \sigma$, so descending on σ lowers a *bound* on
388 the divergence; but a driven system does not relax to $\sigma = 0$. It relaxes to
389 a non-equilibrium steady state at strictly positive entropy production, and
390 the bound then yields only $D_{KL} \leq \Delta t \sigma_{\text{NESS}}$, a positive number. Nothing
391 forces the divergence to zero. The inequality itself is true, is formalized as an

instance obligation, and does real work — combined with Gibbs’ inequality
it gives $\sigma \geq 0$ — but the conclusion drawn from it was not licensed, and
it was that conclusion which supplied this framework’s entire connection to
predictive processing.

The replacement is a published theorem. Still, Sivak, Bell and Crooks
(Still et al., 2012) prove that for a system driven by an environmental signal
the work dissipated over one step of the drive obeys

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (1)$$

The bracket is the *nonpredictive* information: what the structure retains
about the signal’s present that says nothing about the signal’s future. Dis-
sipation is bounded below by it. Memory that fails to predict is not merely
useless to the structure; it is thermodynamic waste, and the waste is quan-
tified in energy.

Three consequences, and they are theorems rather than readings.
The bound is carried in Lean as a class field, `PredictiveDissipation`.
`still_bound (Phase3 PredictiveThermodynamics)`, for the same reason ev-
ery other physical postulate here is: its `axiom` form, quantified over all joint
laws with the dissipated work fixed, is refutable, and `Examples §18.4` refutes
it. What is *derived* is everything downstream.

First, *the bracket is non-negative*, and this is proved rather than assumed.

411 If the structure does not act on its environment, then $X_t \rightarrow S_t \rightarrow S_{t+1}$
 412 is a Markov chain and the data processing inequality gives $I(X_t; S_{t+1}) \leq$
 413 $I(X_t; S_t)$: a system cannot know more about the signal's future than about its
 414 present (`predictiveInfo_le_mutualInfo`). Mathlib carries the Kullback–
 415 Leibler divergence but no mutual information, so the latter is defined here
 416 as the divergence of the joint law from the product of its marginals, which is
 417 the standard definition and makes the library's data processing lemma apply
 418 directly.

419 Second, *the second law follows* instead of being assumed. Dissipation is
 420 bounded below by a quantity now known to be non-negative, so $W_{\text{diss}} \geq 0$
 421 (`dissipatedWork_nonneg`). This mirrors the role Gibbs' inequality plays for
 422 the older bound, with one difference that matters: Gibbs' inequality com-
 423 pares a system to an arbitrary reference distribution, whereas the data pro-
 424 cessing inequality compares the system to its own future, which is a quantity
 425 selection can act on.

426 Third, and this is the point of the replacement, *the inference now runs*
 427 *in the valid direction*. The quantity of interest sits on the small side of
 428 the inequality: $I(X_t; S_t) - I(X_t; S_{t+1}) \leq W_{\text{diss}}/k_B T$. No limit is required.
 429 Any bound on dissipation is immediately a bound on wasted memory, so a
 430 structure held to an energy budget is thereby held to a prediction standard:
 431 this is `nonpredictive_le_dissipation`. In the idealised quasi-static case
 432 $W_{\text{diss}} = 0$, the theorem `predictive_eq_of_no_dissipation` gives that every
 433 retained bit is predictive. This is a weaker statement than the one it replaces,

434 and it is the one that is true.

435 **The hypothesis is load-bearing, and we exhibit what breaks with-**
436 **out it.** The data processing step requires that the structure not write to
437 its environment. In the formalization this is not an assumption stated beside
438 the theorem but the shape of the evolution itself: the signal's step is applied
439 to the pair as the parallel kernel $\text{id} \parallel \kappa$, which acts as the identity on the
440 system coordinate and through κ on the signal coordinate, so no channel
441 from X_t to S_{t+1} exists. **Examples** §18.3 shows the restriction is not decora-
442 tion: a Markov kernel on the *pair* which copies the system's state into the
443 signal carries an independent two-bit law to a perfectly correlated one, rais-
444 ing mutual information from zero to a positive value. An agent that steers
445 its environment can indeed know more about that environment's future than
446 about its present, and Still's bound is not about such an agent.

447 **Both regimes are exhibited, not just described.** **Examples** §18 builds
448 two systems over one correlated two-bit joint law, differing only in the signal's
449 dynamics. When the signal is frozen, every remembered bit is a predictive
450 bit, the nonpredictive information is zero, and the instance is permitted to
451 dissipate nothing — with the memory itself proved strictly positive, so the
452 conclusion is about a system that does remember something. When the signal
453 is redrawn from its stationary law at each step, the predictive information is
454 exactly zero, the *entire* memory is nonpredictive, and the instance is *forced*
455 to dissipate $k_B T$ times its whole mutual information. Establishing that the

456 mutual information of the witness is neither zero nor infinite is where the
457 work of that section lies.

458 **An equality, where the old argument had only a bound.** Kawai,
459 Parrondo and Van den Broeck (Kawai et al., 2007) show that the average
460 dissipated work is exactly $k_B T$ times the relative entropy between the for-
461 ward and time-reversed phase-space densities. Carried here as a predicate on
462 an instance rather than as a further postulate, it replaces the abstract dis-
463 sipation field with a quantity an experiment can estimate — the statistical
464 distinguishability of a recording of the system from the same recording run
465 backwards — and yields the statement that the irreversibility of the dynamics
466 is at least the amount of memory that fails to earn its keep (`nonpredictive_`
467 `le_arrow`).

468 **What this does not establish.** It does not deliver the Free Energy Prin-
469 ciple. Still’s bound says that prediction is thermodynamically favoured; it
470 says nothing about hierarchical generative models, variational free energy,
471 or Bayesian inference, and we do not claim to have derived any of them.
472 The Free Energy Principle (Friston, 2010; Friston & Kiebel, 2009; Hohwy,
473 2013; Clark, 2016; Spisák & Friston, 2026) and the General Resonance The-
474 ory of Consciousness (Hunt & Schooler, 2019) are consonant with what is
475 proved here, which is a weaker relation than derivation and the only one
476 we assert. Nor does it solve the symbol grounding problem (Harnad, 1990).
477 Maximal predictive information is compatible with the structure storing a

478 lossy, compressed or otherwise unrecognisable function of the signal; a state
479 that predicts is not thereby a state that means, and earlier versions of this
480 section overstated the case. What survives of that argument is narrower and
481 still worth saying: whatever internal states a physical structure holds, those
482 that do not help predict its environment are paid for in heat, so the pressure
483 toward states indexed to the environment is energetic rather than semantic.
484 We continue to call the phenomenon *structural resonance*, and there is no
485 teleological drive to “survive” in it: predicting the environment is simply the
486 cheaper of the two ways to hold a memory.

487 **7 Derivation 4: Macroscopic Scaling and Elec-** 488 **trodynamics**

489 To resonate with highly complex environments, the system must scale up,
490 coupling millions of minimal dissipative structures. A highly coupled network
491 cannot allow all of its nodes to reach their lowest energy states simultane-
492 ously. This creates *geometric frustration*, resulting in a jagged phase space
493 of near-equal energy states that provides the physical capacity for complex
494 memory.

495 To manage this frustrated phase space without excessive dissipation, the
496 system undergoes coarse-graining (governed by the Renormalization Group).
497 At the macroscopic scale of biology, emerging paradigms in neuroscience sug-
498 gest these topological constraints manifest as classical continuous electromag-

499 netic fields. Building on seminal EM field theories of consciousness (McFad-
 500 den, 2002; Pockett, 2012) and recent work demonstrating that macroscopic
 501 electric fields (ephaptic coupling) govern representational drift, form memory
 502 networks, and sculpt neural infrastructure (Pinotsis & Miller, 2022, 2023a,b),
 503 we posit that *the continuous macroscopic EM field is the primary dissipative*
 504 *structure interacting with the universe*. Instead of discrete wires, neurons in-
 505 duce fields that shift the membrane potentials of neighboring cells. Modeling
 506 and *in vitro* experiments confirm these endogenous fields are strong enough
 507 to guide network activity and contribute to brain complexity (Goldwyn &
 508 Rinzel, 2016; Ruffini & Santarnecchi, 2020; Cunha & Lima, 2024; Fröhlich &
 509 McCormick, 2010). As proposed by the CEMI field theory, the continuous
 510 electromagnetic field enables spatial integration of information, providing a
 511 physical medium for a unified conscious gestalt (McFadden, 2020).

512 This continuous medium is physically supported by astrocytic syncytia—
 513 non-neuronal networks joined by gap junctions that provide macroscopic cy-
 514 toplasmic continuity for energy and ion flows (Verkhratsky & Nedergaard,
 515 2018). Managing the thermodynamic cost of this state requires sophisticated
 516 micro-bioenergetics. Dynamic mitochondrial networks function as deep-level
 517 Mitochondrial Information Processing Systems (MIPS), transducing biolog-
 518 ical and environmental information to sustain the coherence of the macro-
 519 scopic field (Picard & Shirihi, 2022). Discrete synaptic firings, coupled
 520 with mitochondrial micro-mechanics, generate and sustain the field. Because
 521 the external environment is continuous, achieving maximal thermodynamic

522 structural resonance is optimally realized within this continuous macroscopic
523 medium (van Bree et al., 2025).

524 8 Derivation 5: Global Sections and the Unity 525 of Consciousness

526 The system is now characterized by a continuous macroscopic electrodynamic
527 field generated by millions of frustrated micro-components, forced to resonate
528 with the environment.

529 The Kuramoto model (Kuramoto, 1984) of coupled non-linear oscilla-
530 tors guarantees that, past a critical coupling threshold, these distributed
531 local fields undergo a phase transition into unified synchronization (Bauer et
532 al., 2022). Our Lean proofs (`Phase4_KuramotoDynamics.lean`) confirm this
533 phase-locked state minimizes the Lyapunov potential of the system, in the
534 precise sense discussed in Section 2.1: the functional minimized is the cou-
535 pling potential read in the frame rotating with the natural frequency, which
536 `Phase4_RotatingFrame.lean` shows is the honest reduced description of a
537 system of identical oscillators.

538 This phase transition mathematically glues local synchronized states into
539 a singular *Global Section* of a probability sheaf (`Phase5_GlobalSection.lean`).
540 The EM field acts as a single, indivisible topological space, providing the
541 *Unity of Consciousness*—a necessary condition for integrated experience that
542 solves the binding problem. While Global Workspace Theory (Baars, 1997;

543 Dehaene & Marti, 2014) models this unity as a broadcast architecture and
 544 Integrated Information Theory (Tononi & Koch, 2016) quantifies it via Φ ,
 545 our derivation provides a continuous physical mechanism. The contrast is
 546 worth drawing carefully, because both theories are addressing the same ex-
 547 planandum from different directions.

548 We state the scope of the Lean result precisely, because it is narrower
 549 than the prose above. `ThermodynamicCover` demands that the cover’s phase
 550 field minimize the reduced Kuramoto potential, and `potential_min_iff_`
 551 `phase_locked` characterizes those minimizers exactly: they are the phase-
 552 locked configurations, in both directions. Every instance is therefore already
 553 phase-locked (`ThermodynamicCover.phase_locked`), and the class’s single
 554 condition on its local data — that patches at a common phase agree where
 555 they overlap (`section_agrees_of_phase_eq`) — makes the family of local
 556 sections compatible. `probability_glue_unique`, which is the sheaf property
 557 of the probability presheaf and contains no physics at all, then glues that
 558 family into exactly one global section.

559 An earlier version of this structure proved less than the prose claimed,
 560 and we record the correction because it changed what the theorem says
 561 rather than how it is stated. `LocalSectionSynchronization` used to carry
 562 a *global* section as a class field, together with a second field declaring every
 563 local section to be a restriction of it. The global object therefore existed
 564 before the sheaf condition was ever consulted, and `global_section_from_`
 565 `thermodynamics` could do no more than rule out competitors to an object

566 the instance had supplied. Both fields are gone. Every field of the class
 567 now mentions a single patch or a single overlap; the global section is con-
 568 structed (`ThermodynamicCover.invariantMeasure`); and the removed field
 569 survives as a *theorem* about that construction (`ThermodynamicCover.sync_
 570 to_section_eq`), so the local sections are restrictions of one common measure
 571 because the sheaf condition says so, not because an instance was required to
 572 name one.

573 `Examples.lean` §14 is the witness that the difference is not notational.
 574 Two patches carry two different mass profiles, $(2, 1, 7)$ and $(5, 1, 3)$ across
 575 the three sites, agreeing only at the site they share, where both put mass 1
 576 — which is what discharges the overlap condition, by computing two num-
 577 bers rather than by construction. The section the theorem returns has pro-
 578 file $(2, 1, 3)$: it is neither of the two profiles (`gluedW_ne_leftW`, `gluedW_ne_
 579 rightW`), and neither of them restricts correctly to both patches (`leftW_
 580 glues_nothing`, `rightW_glues_nothing`). The global state carries informa-
 581 tion that no single patch had. `Examples.lean` §13, built the older way
 582 through the constructor `LocalSectionSynchronization.ofInvariantMeasure`,
 583 is retained alongside it for the negative result it carries: its phase field is not
 584 the constant function — the second patch a full turn ahead of the first — and
 585 by `ThermodynamicCover.phase_locked` that is as far from constant as any
 586 instance can get, since no instance can carry phases differing by anything a
 587 measure is allowed to see.

588 Two hypotheses carry physical content, and until recently each was an

obligation on every instance. One still is: that synchronized patches agree
 on their overlaps is a modelling assumption, not a consequence of any dy-
 namics we formalize. The other, `thermodynamic_equilibrium`, asks that the
 cover already sit at the potential minimum, and no theorem says an *arbi-*
trary system’s dynamics takes it there. “A thermodynamic phase transition
 into unity” is therefore still doing causal work that no general Lean result
 supports, though it is no longer entirely without one: `Examples.lean` §15
 exhibits a two-oscillator trajectory that starts a quarter turn out of phase
 and converges to the global minimum of the dynamic potential, with the or-
 der parameter tending to 1. That trajectory is not an isolated construction:
`kuramoto_cauchy_problem` proves the Kuramoto initial value problem well
 posed on all of $\mathbb{R}$, so through every initial state there passes exactly one tra-
 jectory, and the exhibited one is the only solution through its own. Nor is its
 settling down peculiar to it: along every trajectory of a zero-frequency sys-
 tem the dynamic potential converges, and the total dissipation $\int_0^\infty \sum_i \dot{\theta}_i^2$ is
 finite and equal to the potential’s total drop (`dynamic_potential_tendsto`,
`dissipation_integral_tendsto`). The motion also stops: a finite integral
 does not by itself force its integrand to zero, but the dissipation rate is
 Lipschitz in time, so Barbalat’s lemma applies, and every trajectory satis-
 fies $\sum_i \dot{\theta}_i^2 \rightarrow 0$ (`velocity_sq_tendsto_zero`; Mathlib has no Barbalat, so it
 is proved here as `tendsto_zero_of_lipschitz_of_integral_le`). That last
 step — locating where the motion stops — is now taken, though condition-
 ally, and by a different route than we had recorded. We had marked it as

612 blocked on the LaSalle invariance principle, which Mathlib does not have,
 613 and on the absence of a compact invariant set in the state space $V \rightarrow \mathbb{R}$.
 614 That was the wrong diagnosis. LaSalle is the tool for a system with a de-
 615 creasing functional and nothing else; this one has more. The flow is a gra-
 616 dient flow of the dynamic potential, so $\dot{V} = -\sum_i \dot{\theta}_i^2$ is an identity rather
 617 than an inequality; the minimisers of V are characterised exactly; and on
 618 the region where every pair of phases lies within a quarter turn a Polyak-
 619 Łojasiewicz estimate holds, $2a^2W \leq (\sum_{ij} A_{ij}) \sum_i \dot{\theta}_i^2$, where W is the poten-
 620 tial's excess above its minimum and $a > 0$ is a lower bound for the coupling
 621 (`lojasiewicz_estimate`). Its proof is two extremal indices and the iden-
 622 tity $\sin^2 D = (1 - \cos D)(1 + \cos D)$: the site carrying the largest phase is
 623 pulled backwards at rate at least $a \sin D$, where D is the spread, while the
 624 excess is at most $\frac{1}{2}(\sum_{ij} A_{ij})(1 - \cos D)$. The compactness objection dissolves
 625 as well, because the invariant region is produced by the Lyapunov function
 626 rather than assumed: a single pair contributes $\frac{1}{2}a(1 - \cos)$ to the excess, so
 627 an excess below $a/2$ forces every pairwise cosine positive for all time, and
 628 continuity on the connected half-line then prevents any pair from chang-
 629 ing branch (`spread_le_of_small`). Descent plus the estimate integrates to
 630 $W(t) \leq W(0)e^{-ct}$ with $c = 2a^2 / \sum_{ij} A_{ij}$ (`excess_decay`), the rate makes each
 631 $|\dot{\theta}_i|$ exponentially small and hence integrable on $[0, \infty)$, and a trajectory with
 632 integrable velocity converges. The conclusion is `kuramoto_tendsto_global_`
 633 `minimum`: if the phases start within a quarter turn of each other and the initial
 634 excess is below $a/2$, the trajectory converges to a phase-locked configuration

635 which is a *global* minimiser of the dynamic potential, with order parameter
 636 $r^2 = 1$. That is the configuration `thermodynamic_equilibrium` postulates,
 637 reached rather than assumed. `Examples.lean` §17 fires it on three sites with
 638 initial data $(0, 0, \frac{1}{2})$, where the Kuramoto system has no closed-form solu-
 639 tion: the trajectory comes from the existence theorem and its limit from the
 640 convergence theorem, so neither is written down.

641 `Phase5_EquilibriumBridge.lean` closes the gap that remained between
 642 the two results. `ThermodynamicCover.ofConvergentTrajectory` takes a
 643 cover whose phase field is the limit of such a trajectory and returns an in-
 644 stance whose `thermodynamic_equilibrium` field is discharged *by that con-*
 645 *vergence*; the cover's coupling matrix is the coupling matrix of the dynamics
 646 rather than a separately chosen one, so the system whose equilibrium the
 647 class asserts and the system whose trajectory is run are the same system.
 648 `Examples.lean` §17.1 is the resulting witness: three patches over the three-
 649 site substrate, each carrying its own mass profile on its own open set, above a
 650 phase field that no line of the construction writes down. The trajectory that
 651 reaches it starts at $(0, 0, \frac{1}{2})$, which is provably not phase-locked (`trioCover_`
 652 `start_not_locked`), so on this cover the configuration Derivation 5 requires
 653 is arrived at rather than posited. The state the three patches glue to has
 654 profile $(2, 1, 3)$ and is none of the three profiles the instance carries (`trioW_`
 655 `ne_glued`), so one cover now carries both the emergence reading of the gluing
 656 and the derived reading of the equilibrium. What is still assumed on it is the
 657 overlap agreement, and nothing else.

658 The two hypotheses cannot be dropped, and this is the honest scope of
 659 the result. Splay and twisted configurations are equilibria of the same flow,
 660 so “every trajectory reaches the unified state” is false as a general statement,
 661 and the initial-data condition is the standard arc condition under which the
 662 true statement holds. What is claimed is therefore conditional — unity is
 663 reached from configurations already sufficiently coherent — and not that an
 664 arbitrary initial state is carried into unity. What Lean establishes about the
 665 gluing itself is the other half: *given* locally synchronized patches, a global
 666 state exists, is unique, and is produced by the gluing rather than presupposed
 667 by it.

668 Integrated Information Theory begins from phenomenological axioms and
 669 derives a scalar Φ measuring irreducibility of a system’s cause-effect structure,
 670 with the exclusion postulate selecting a single maximum- Φ complex as the
 671 locus of experience. Our sheaf-theoretic global section plays a structurally
 672 analogous role — it too picks out a distinguished unified object — but it
 673 is arrived at from the opposite end: not postulated from the character of
 674 experience, but forced by a thermodynamic phase transition in a physical
 675 field. This buys two things IIT does not have. First, the unity condition
 676 is checkable from field dynamics rather than requiring the computation of
 677 Φ , which is intractable for systems of realistic size and whose definition has
 678 undergone repeated revision (Tononi & Koch, 2016). Second, exclusion falls
 679 out rather than being postulated: gluing either succeeds on a given cover or
 680 it does not, so there is no separate principle needed to rule out overlapping

conscious subsystems. It also inherits IIT’s exposures. Critics have argued that Φ can be high in systems with no plausible claim to experience, such as large feedforward-free grids (Aaronson, 2014); our criterion is vulnerable to the mirror objection, since a physical system could in principle satisfy the gluing condition without being cognitively interesting. We take this seriously rather than dismissing it: our reply is that gluing is necessary but not sufficient, and that the topological boundary of Derivation 1 and the reflexive structure of Derivation 6 are additional required conditions. IIT, by contrast, treats Φ as both necessary and sufficient.

Global Workspace Theory faces a complementary difficulty. It specifies a functional architecture — a limited-capacity workspace broadcasting to specialized modules — without specifying what physical property of broadcasting makes the broadcast content conscious, leaving the theory open to the charge that it explains access consciousness while leaving phenomenal consciousness untouched (Block, 1995). Our framework can be read as supplying the missing physical substrate for the workspace: the continuous macroscopic field *is* the medium in which integration is realized, and the $K_c = 2D$ threshold gives a physical condition for ignition rather than a functional stipulation of it. Where GWT is agnostic about implementation, we make a falsifiable commitment — unity requires a continuous physical medium, which predicts that a functionally equivalent discrete broadcast architecture would not be conscious. We regard this as a genuine cost of the position, not a free advantage: it is the point at which our framework departs most sharply from

704 computational functionalism and where it is most readily falsified. Contrast
 705 this derivation with panpsychism: consciousness is not assumed as a funda-
 706 mental property of matter. It requires the initial existence of a topological
 707 defect to establish a bounded “inside” (Derivation 1), followed by a thermo-
 708 dynamic phase transition into unity that occurs when integrated spatial cou-
 709 pling exceeds the thermodynamic noise threshold ($K_c = 2D$). Without both
 710 the foundational topological boundary and the subsequent mathematically
 711 defined phase transition, the system remains a fragmented, non-conscious
 712 collection of micro-states. A unified field of perception, however, is not yet
 713 a “Self.”

714 **9 Derivation 6: Reflexive Topology and the** 715 **Emergence of the Self**

716 While a global section provides Unity, a “Self” requires reflexivity. The
 717 unified EM field is a dissipative structure that must resonate with external
 718 perturbations while managing its own internal phase transitions. Time evo-
 719 lution naturally drives the field to minimize thermodynamic friction with its
 720 own internal state-changes.

721 This mandates *auto-resonance*: the field must predict itself. The “Self”
 722 emerges as a higher-order topological feature when the internal geometry
 723 folds to include a low-dimensional avatar of its own computational and causal
 724 boundary within its predictive model (`Phase6_ReflexiveTopology.lean`)

725 (Levin, 2019). The system is unified, predictive, and explicitly maps its
 726 own distinct existence from the environment. This architecture provides the
 727 physical foundation for a self-aware consciousness, bounded by the continuous
 728 measure thermodynamics of Landauer erasure (`Phase3_MeasureThermodynamics.lean`).
 729 We are explicit about what carries the weight here. `reflexive_topology_implies_self`
 730 is the Banach fixed-point theorem with the physical commitments in its hy-
 731 potheses: the space of global sections is assumed nonempty, metric and com-
 732 plete, and the predictive map is assumed to contract by $1/2$. `Examples.lean` §10
 733 discharges all four on the three-site substrate of Derivation 5, and the metric
 734 is no longer a placeholder. The section first pierces the sheafification: on
 735 a discrete substrate every point has a smallest open neighbourhood, so the
 736 mass a germ carries at a point is well defined, and `massEquivOn` proves that
 737 over *any* open the sections of the probability sheaf *are* the finite measures
 738 on that open, read as densities against counting measure — the global case
 739 being $U = \top$. Mathlib supplies no inverse to this sheafification — its `TopCat`
 740 sheafification has neither an adjunction nor an `isIso_toSheafify` — so the
 741 correspondence had to be constructed. The metric on sections is then the
 742 uniform distance between the densities of the measures they glue to — equiv-
 743 alent to their total variation on a finite substrate, though not equal to it —
 744 and it is proved complete by exhibiting the limit section of a Cauchy sequence
 745 of densities, not by the degenerate route of showing Cauchy sequences are
 746 eventually constant. The self-prediction map is the relaxation “average the
 747 current field with the baseline,” and we prove it contracts distances by *ex-*

748 *actly* one half, that it is not constant, and that its fixed point is unique. An
 749 earlier version of this witness used the 0/1 metric, under which every con-
 750 traction is constant; that theorem is kept in the source as the record of why
 751 the earlier witness was empty. A separate gap, closed at the same time, con-
 752 cerned the avatar rather than the fixed point. **ReflexiveBoundary** carries a
 753 field **auto_resonance** sending the global state to a section over the avatar re-
 754 gion, and nothing constrained it: a constant function was a legal avatar, and
 755 **reflexive_topology_implies_self** consumes only the predictive model, so
 756 no theorem in the development mentioned the field the avatar is supposed to
 757 be reading. The constraint is now stated as a predicate rather than added
 758 to the structure — **IsRestrictionResonance**, that the avatar’s state is the
 759 presheaf restriction of the global state — and two theorems say why it is
 760 the right one. In the positive direction, if a family of resonant avatars *cov-*
 761 *ers* the substrate then the global state is determined by their local readings
 762 alone (**eq_of_avatar_eq**): unity is not assumed alongside the avatars but
 763 recovered from them, by the same sheaf condition that Derivation 5 uses.
 764 In the negative direction, replacing any avatar by a constant leaves a legal
 765 **ReflexiveBoundary** with the same region, the same predictive model and
 766 the same Self (**self_of_constResonance**), so the fixed-point theorem cannot
 767 see the difference and the predicate is doing the work rather than restating
 768 it. Both halves are witnessed on the three-site substrate, where the avatar
 769 region provably separates two states of the field, so the condition is a con-
 770 straint there and not a formality. What is still not established is that any field

771 dynamics produces this particular map or performs the restriction, or that
772 the contraction constant $1/2$ is read off an entropy production rate rather
773 than built into the map’s definition. We also note a negative result found
774 in the course of this work: the Lévy–Prokhorov metric, which the natural
775 weak-convergence reading would suggest, does *not* support this argument on
776 a discrete substrate, because the ε -thickening of a set is the set itself for $\varepsilon < 1$
777 and the relaxation map is consequently not a Lévy–Prokhorov contraction.

778 10 Derivation 7: Continuous Stochastic Ther- 779 modynamics and Field Theory

780 To move beyond finite graph analogies, we model the macroscopic electro-
781 dynamic field as a *Continuous Neural Field*. True ephaptic coupling is a
782 continuous spatial phenomenon, not a discrete network. We extend the
783 discrete Kuramoto model to a continuous spatial coupling kernel $K(x, y)$
784 (`Phase8ContinuousField.lean`).

785 Subject to thermodynamic noise D , the system undergoes non-equilibrium
786 stochastic thermodynamics. Rather than a static Landauer bound, the field
787 dynamics minimize an entropy production functional σ :

$$\sigma = \int \frac{1}{D} \left(\omega(x) + \int K_t(x, y) \sin(\theta_y - \theta_x) dy \right)^2 dx \quad (2)$$

788 Biological fields dynamically deform their spatial topology $K_t(x, y)$ to un-

789 dergo gradient descent on this entropy production functional.

790 That gradient descent is a theorem rather than a modelling assertion,
791 for finite substrates. Reading σ as a functional of the coupling kernel at
792 fixed phases, `hasFDerivAt_sigmaOfKernel` computes its Fréchet derivative
793 in closed form,

$$\frac{\partial \sigma}{\partial K(a, b)} = \frac{2}{D} \left(\omega(a) + \sum_y K(a, y) \sin(\theta_y - \theta_a) \right) \sin(\theta_b - \theta_a), \quad (3)$$

794 and `sigmaOfKernel.eq_entropy_production_rate` identifies the functional
795 being differentiated with `entropy_production_rate` itself whenever the sub-
796 strate carries counting measure. `structural_resonance_decreases_entropy_production`
797 then concludes that σ is non-increasing along any coupling trajectory relax-
798 ing down its own gradient at a positive rate c . Earlier drafts proved this
799 only for an abstract functional S on an abstract inner-product space with
800 the gradient supplied as data; the identification of that S with σ was infor-
801 mal, and Table 1 flagged the row accordingly. Two limitations stay explicit,
802 and the first is narrower than we previously reported. The derivation is for
803 *finite* substrates. We had recorded the obstacle to the continuum case as
804 differentiation under the integral sign with respect to the kernel; that was
805 a misdiagnosis. σ is not a general functional that happens to be written as
806 an integral — it is a quadratic functional of an *affine* image of the kernel,
807 since the drift is a fixed ω plus a linear operator applied to K , and σ is a
808 constant times the squared norm of the drift. A squared norm is Fréchet

809 differentiable on any real inner-product space and an affine map is its own
 810 derivative, so the composite differentiates by the chain rule with no limiting
 811 argument. `hasFDerivAt_quadratic_of_affine` states that in full generality,
 812 and `hasFDerivAt_sigmaOfKernel_of_operator` re-derives the finite gradient
 813 from it, showing that the closed form displayed above is a convenience rather
 814 than the content. What the continuum case still needs is therefore one object:
 815 the drift map $K \mapsto (x \mapsto \int K(x, y) \sin(\theta_y - \theta_x) dy)$ as a bounded operator
 816 $L^2(\mu \otimes \mu) \rightarrow L^2(\mu)$, which is a Cauchy–Schwarz estimate — the integral ker-
 817 nel is bounded by 1, so on a finite measure space the operator norm is at most
 818 $\mu(M)^{1/2}$. We have not built it, because it requires constructing a continu-
 819 ous linear map between L^p spaces and we have not done that bookkeeping;
 820 we record the corrected scope because “construct one bounded operator”
 821 and “differentiate under the integral sign” are different sizes of task. The
 822 second limitation is unchanged: the phase field is held fixed throughout, so
 823 this is plasticity of the coupling at frozen phases, not joint (θ, K) dynamics.
 824 **Examples.lean** §5 exercises the whole chain on a two-site substrate whose
 825 coupling relaxes as e^{-t} and whose entropy production decays as e^{-2t} , con-
 826 firming that the gradient is not identically zero and that the descent is strict
 827 rather than vacuously monotone.

828 A second result about σ concerns where the descent is headed, and it has
 829 been strengthened since the last draft in a way that also corrects an estimate
 830 we had recorded. That the phase-locked field minimizes σ was proved only
 831 against competitors sharing its spatial mean drift — a restriction that left

832 the theorem close to Jensen’s inequality and no more — and we had judged
 833 that lifting it would require modelling how the mean drift varies with the
 834 competitor. That judgement was wrong. For a *reciprocal* kernel, one with
 835 $K(x, y) = K(y, x)$, the restriction is not a restriction: the coupling integrand
 836 $K(x, y) \sin(\theta_y - \theta_x)$ is antisymmetric under exchanging its two sites, because
 837 K is symmetric while $\sin$ is odd, so Fubini identifies its double integral with
 838 its own negative and the total coupling contribution vanishes. The total drift
 839 $\int (\omega + \text{coupling})$ is therefore $\int \omega$ whatever the phases are (`total_drift_eq_of_`
 840 `symm`): coupling moves drift between sites but neither creates nor destroys it.
 841 Evaluating that conserved quantity at the phase-locked configuration, where
 842 the drift is constant, pins its value, and the mean-matching hypothesis follows
 843 for every competitor. `phase_locked_minimizes_entropy_of_symm` is the un-
 844 restricted statement. Reciprocity is the physically intended condition — an
 845 ephaptic field acts on both sites of a pair alike, and it is the same symmetry
 846 `ThermodynamicCover.A_symm` already imposes in Derivation 5 — but it is
 847 not a formality, and we prove that rather than assert it. `Examples.lean` §16
 848 exhibits a *one-way* coupling on two sites, in which a feels b and b does not
 849 feel a , whose phase-locked field is strictly beaten: displacing the second phase
 850 to $-\pi/2$ cancels the first site’s natural drift while adding none at the sec-
 851 ond, halving σ . So “phase-locking minimizes entropy production” is false for
 852 general kernels and true for reciprocal ones, and the hypothesis cannot be
 853 removed by working harder.

854 We define the critical coupling threshold $K_c = 2D$. The phase transition

855 to the unified “Global Section” occurs when the mean-field spatial coupling
856 exceeds this thermodynamic noise threshold, linking the emergence of Unity
857 to physical temperature and noise.

858 Earlier drafts left $K_c = 2D$ as a bare stipulation validated only by
859 simulation. `Phase8_SelfConsistency.lean` now proves the half of it that
860 is ordinary real analysis. For the noisy mean-field Kuramoto model with
861 identical frequencies, the stationary density at order parameter r is the
862 von Mises density $\rho(\theta) \propto e^{(Kr/D)\cos\theta}$, and self-consistency requires $r =$
863 $R(K, r) := I_1(Kr/D)/I_0(Kr/D)$. Mathlib has no modified Bessel functions,
864 so we formalize that ratio directly as a ratio of integrals over $[-\pi, \pi]$. In-
865 tegration by parts with $u = \sin$, $v = -e^{a\cos}$ gives an exact identity with
866 no boundary term, $\int \cos\theta e^{a\cos\theta} d\theta = a \int \sin^2\theta e^{a\cos\theta} d\theta$, so R is a times
867 the mean of $\sin^2$ under the von Mises weight. Bounding that mean by 1
868 would put the threshold at D ; the true threshold is $2D$, and the entire fac-
869 tor of two is the sharper bound $\mathbb{E}_a[\sin^2\theta] \leq 1/2$, equivalently $I_2(a) \geq 0$.
870 We prove it elementarily, by folding $[-\pi, \pi]$ onto $[0, \pi/4]$ in two reflections
871 ($\theta \mapsto \pi - \theta$, then $\theta \mapsto \pi/2 - \theta$), after which the integrand is a product
872 of $\cos 2\theta \geq 0$ with $2 \cosh(a \cos \theta) - 2 \cosh(a \sin \theta) \geq 0$ — the second factor
873 because $0 \leq \sin \theta \leq \cos \theta$ there and $\cosh$ increases in $|\cdot|$. The conclusion,
874 `subcritical_fixed_point_eq_zero'`, is that for $K < K_c$ the incoherent state
875 $r = 0$ is the *only* non-negative solution of the self-consistency equation; $r = 0$
876 is exhibited as a solution, so the uniqueness claim is not vacuous. Three fur-
877 ther facts about the threshold have since been proved, and they narrow the

878 gap between this row and a full “Theorem” to a single named statement.
 879 First, the fold argument gives a *strict* inequality for $a > 0$ (`vonMisesC2_`
 880 `pos`, hence $R(a) < a/2$), which settles the threshold itself: at $K = K_c$ the
 881 self-consistency map is tangent to the diagonal at the origin but falls strictly
 882 below it thereafter, so $r = 0$ remains the only solution and the bifurcation
 883 happens strictly *after* K_c , not at it (`fixed_point_eq_zero_of_le_critical`).
 884 Second, factoring $R(a) = a E(a)$ shows that a non-zero r solves the equation
 885 exactly when $E(Kr/D) = D/K$ (`coherent_iff_sRatio_eq`), so the coher-
 886 ent branch is a level set of the single function $E(a) = \mathbb{E}_a[\sin^2 \theta]$. Third,
 887 and using only $E < 1/2$ together with compactness, every coherent solution
 888 tends to zero as $K \downarrow K_c$ (`coherent_branch_continuous_at_threshold`): the
 889 branch emerges continuously from the incoherent state rather than appearing
 890 at a finite distance from it, which is what makes the transition supercritical
 891 in the technical sense. That statement quantifies over *every* solution, so it
 892 does not presuppose the uniqueness it does not have. The level-set form
 893 also says exactly what uniqueness is: the injectivity of E on $(0, \infty)$. That
 894 is now proved, in the stronger form that E is strictly decreasing on $[0, \infty)$
 895 (`vonMisesSRatio_strictAntiOn`), and by an argument that avoids the route
 896 we had expected to need. Differentiating the exponential family in its natu-
 897 ral parameter gives $E'(a) = -\text{Cov}_a(\cos^2 \theta, \cos \theta)$, which costs differentiation
 898 under the integral sign and leaves a covariance that is not sign-definite point-
 899 wise: symmetrising with an independent copy gives $\frac{1}{2}\mathbb{E}[(X - X')^2(X + X')]$
 900 with $X = \cos \theta$, an integrand changing sign with $X + X'$. Nothing in the

901 proof differentiates anything. The target is kept as the two-point inequality
 902 $C_2(a)Z(b) < C_2(b)Z(a)$ for $a < b$, and two moves settle it. *Fold first*: the
 903 two reflections that proved $I_2 \geq 0$ carry both integrals onto $[0, \pi/2]$, replacing
 904 $e^{a \cos \theta}$ by $W(a, \theta) = 2 \cosh(a \cos \theta)$ and confining $\cos \theta$ to $[0, 1]$, after which
 905 $\cos 2x - \cos 2y$ and $W(b, x)W(a, y) - W(a, x)W(b, y)$ have the same sign, both
 906 being governed by $\cos x$ against $\cos y$. This is the symmetrisation $X \mapsto -X$
 907 that the covariance form needed the tilt to perform, done once at the start; its
 908 analytic content is one identity, $2 \cosh X \cosh Y = \cosh(X+Y) + \cosh(X-Y)$
 909 with $\cosh$ increasing in $|\cdot|$. *Then cross*: instead of integrating that inequality
 910 over a square, let α be the mean of $\cos 2x$ under $W(a, \cdot)$ and $x_0 \in [0, \pi/2]$ a
 911 point where $\cos 2x_0 = \alpha$, supplied by the intermediate value theorem; eval-
 912 uating at $y = x_0$ and integrating in x alone gives the result, since the term
 913 carrying $W(b, x_0)$ vanishes by the choice of α . No product measure and no
 914 Fubini appear. Run with $\cos \theta$ in place of $\sin^2 \theta$ the same argument needs no
 915 fold — $R(a) = \mathbb{E}_a[\cos \theta]$ is the mean of the variable the family is tilted by —
 916 and gives **besselRatio_strictMono**: R is strictly increasing on all of $\mathbb{R}$, the
 917 monotonicity of the Bessel ratio our earlier drafts recorded as unproved.

918 The converse half is **supercritical_fixed_point_exists**: for $K > K_c$
 919 there is an r with $0 < r \leq 1$ solving $r = R(K, r)$. We had expected
 920 this to require a second-order expansion of the Bessel ratio, and it does
 921 not. Writing $R(a) = a E(a)$ with $E(a) = \mathbb{E}_a[\sin^2 \theta]$ — the same factoriza-
 922 tion the integration by parts already supplies — the only facts needed are
 923 that E is continuous and that $E(0) = 1/2$, the latter because the weight

924 is constant at $a = 0$. Since $K > 2D$ makes D/K strictly less than $1/2$,
 925 continuity gives $E > D/K$ on a neighbourhood of the origin, so the map
 926 overshoots, $R(K, r) > r$, for all small $r > 0$; at the other end $R \leq 1$ al-
 927 ways, because R is a mean of $\cos \theta$, so the map undershoots at $r = 1$.
 928 The intermediate value theorem closes the gap. Continuity of E is Math-
 929 lib's continuity of a parametric interval integral with jointly continuous in-
 930 tegrand, so no dominated-convergence argument is written by hand, and
 931 the existence proof never uses monotonicity of R : the fixed point is pro-
 932 duced by the intermediate value theorem, not by iterating the map. That
 933 R is in fact strictly increasing is proved below, and no threshold result de-
 934 pends on it. `critical_coupling_is_threshold` packages the two halves,
 935 and `exhibits_phase_transition_coherent` applies the supercritical half to
 936 any substrate satisfying the predicate of the previous paragraph — so ex-
 937 ceeding K_c now implies the existence of a coherent stationary order pa-
 938 rameter rather than merely comparing against a named number. With
 939 uniqueness available both are sharpened — `supercritical_fixed_point_`
 940 `existsUnique`, `supercritical_solution_set` (the non-negative solutions
 941 are exactly $\{0\}$ below threshold and $\{0, r\}$ above it) and `exhibits_phase_`
 942 `transition_unique_coherent` — and the two monotonicities together fix
 943 the shape of the branch: for $K_1 < K_2$, $E(a_i) = D/K_i$ with E decreasing
 944 puts the concentrations in the order $a_1 < a_2$, and $r_i = R(a_i)$ with R increas-
 945 ing carries that to $r_1 < r_2$ (`coherent_branch_strictMono`). The coherent
 946 order parameter grows strictly with the coupling, which with the continuity

947 at threshold is the qualitative bifurcation diagram.

948 One more step makes the equation mean in Lean what it means on paper.
949 As a definition, `selfConsistency K D r` is just $R(Kr/D)$ — a function of
950 r with a suggestive name, and nothing in the source says it has anything
951 to do with an order parameter. `Phase8_SelfConsistency §8` supplies the
952 identification. `vonMisesDensity` is the normalized weight, proved to be a
953 genuine probability density on $[-\pi, \pi]$; `circularOrderParameter` is the av-
954 erage of $e^{i\theta}$ against a density, the continuum analogue of the finite-system
955 `order_parameter_complex` of `Phase4`; and that average is proved to be ex-
956 actly $R(a)$ — real, because the density is even, so the mean direction van-
957 ishes. `fixedPoint_iff_selfReproducing` then states the equivalence the
958 name promises: r solves $r = R(K, r)$ if and only if the von Mises density
959 it induces has order parameter r . Read through this, the existence theorem
960 says that above K_c there is a stationary density whose own order parameter
961 is positive, and the solution $r = 0$ is identified with the uniform density,
962 whose order parameter is zero.

963 What remains not established, and we separate it carefully. The sta-
964 tionary density is an *input*: deriving it from the Langevin dynamics $d\theta =$
965 $(\omega + K \text{mean-field}) dt + \sqrt{2D} dW$ needs the Fokker–Planck operator, existence
966 and uniqueness of its stationary solution, and a spectral stability argument,
967 none of which Mathlib has. Of the coherent branch we prove existence,
968 uniqueness, continuous emergence at the threshold and strict growth in K ,
969 but not that it is the dynamically selected one: nothing in that file men-

970 tions a trajectory, so stability and selection are not statements it can make.
 971 And while `selfConsistency` is now connected to the order parameter of
 972 a *density*, it is connected to no *trajectory*: `circularOrderParameter` is an
 973 integral, `order_parameter_complex` is an average over finitely many oscilla-
 974 tors, and relating them is the mean-field limit — propagation of chaos for
 975 the finite Kuramoto system — which is a research programme rather than
 976 a lemma. That single gap is now the whole of what separates our threshold
 977 predicate from a statement about dynamics. Table 1 records the row ac-
 978 cordingly as a theorem about the self-consistency equation from an assumed
 979 density, not as a derivation of the threshold from the dynamics. Perceptual
 980 anomalies such as visual snow can be modeled as boundary conditions where
 981 microscopic thermal noise (D) fails to be fully coarse-grained away by the
 982 continuous field, leaking across the K_c threshold into the macroscopic gestalt.

983 Clinical evidence for this phase transition is observed during general anes-
 984 thesia. Work by Miller and colleagues (Bastos & Miller, 2021) demonstrates
 985 that anesthetics like propofol induce unconsciousness not by suppressing neu-
 986 ral firing, but by disrupting the phase alignment of high-frequency cortical
 987 rhythms. Destroying the dynamic stability of these rhythms breaks the struc-
 988 tural resonance of the continuous field, forcing the integrated spatial coupling
 989 below the critical threshold K_c and fracturing the unified Global Section back
 990 into disjoint, unconscious local fields.

991 11 The Sleep-Inertia Timescale Mismatch

992 Most of what this framework asserts is shared with some rival: that corti-
993 cal synchrony matters, that dissipation constrains cognition, that unity is a
994 physical rather than a computational property. One prediction is not shared,
995 and it follows from the framework’s specific commitment — that the medium
996 carrying the unified state is a continuous field whose effective coupling de-
997 pends on the physical geometry of the extracellular space, not on the rhythms
998 alone. Where that commitment differs from a purely electrical account, it
999 differs in an observable way, and we set out the difference precisely enough
1000 to be wrong.

1001 The daily loss of consciousness during sleep is modeled here as an active
1002 physical disruption of the medium rather than a quiescence of the oscillators.
1003 During sleep, astrocytic shrinkage expands the extracellular space, allowing
1004 an influx of cerebrospinal fluid (CSF) to flood the brain for metabolic waste
1005 clearance (Xie & Nedergaard, 2013). This influx alters the spatial medium
1006 itself, disrupting the continuous structural resonance and preventing the co-
1007 herent sheaf-theoretic gluing that conscious unity requires on this account.
1008 The oscillators need not stop for the field to fragment; the geometry they are
1009 coupled through is what changes.

1010 The prediction concerns sudden awakenings. When jolted awake, the lo-
1011 cus coeruleus floods the cortex with noradrenaline. High-frequency electrical
1012 rhythms attempt to phase-align in milliseconds (Bastos & Miller, 2021), but

1013 the noradrenaline-driven astrocytic constriction required to squeeze the ex-
1014 cess CSF out of the interstitial pathways operates on a timescale of seconds
1015 to minutes (Xie & Nedergaard, 2013) — three to six orders of magnitude
1016 slower. During that lag the electrical oscillators are synchronizing across a
1017 waterlogged topology, and the fragmented structural resonance is what is ex-
1018 perienceed as grogginess. The unified Global Section re-establishes only when
1019 astrocytic volume finishes matching the state of the electrical rhythms.

1020 **Why this implies a delayed sigmoid.** The two timescales separate
1021 cleanly, and that separation is what makes the prediction sharp rather than
1022 merely qualitative. Write $K(t)$ for the effective coupling of the continuous
1023 field, which on this account is a monotone function of the reconstricting
1024 extracellular geometry, and let $r(t)$ be the order parameter of the phase
1025 field. The electrical variable is fast: at every instant the oscillators are al-
1026 ready at the configuration appropriate to the current K , so r tracks the
1027 equilibrium branch $r(K)$ rather than relaxing on its own. The slow vari-
1028 able $K(t)$ then carries the whole time course, and it relaxes over 10^2 – 10^3
1029 seconds. What Derivation 7 establishes about that branch is exactly what
1030 the functional form needs, and each of the three pieces is a theorem rather
1031 than a picture: at or below the threshold the incoherent solution $r = 0$ is the
1032 *only* non-negative solution of the self-consistency equation (`fixed_point_eq_`
1033 `zero_of_le_critical`); above it the coherent solution exists and is unique
1034 (`supercritical_fixed_point_existsUnique`); it emerges *continuously* at

1035 K_c rather than by a jump (`coherent_branch_continuous_at_threshold`);
 1036 and it is strictly increasing in K (`coherent_branch_strictMono`). A jump
 1037 discontinuity at threshold would predict a step rather than a sigmoid, and
 1038 a non-monotone branch would predict overshoot; neither is what the mathe-
 1039 matics gives. Composing a monotone relaxation $K(t)$ with a branch that is
 1040 identically zero below K_c , rises continuously from zero at K_c , and saturates
 1041 above it yields a curve with three phases: a genuine *latency* while $K(t) < K_c$,
 1042 during which recovery is not merely slow but absent; a continuous rise as the
 1043 threshold is crossed; and a saturation as $K(t)$ reaches its waking value. That
 1044 is a delayed sigmoid, and the delay is not a fitting parameter — it is the time
 1045 the slow variable takes to reach K_c .

1046 **What the alternative predicts.** An account on which grogginess is the
 1047 electrical realignment itself — the rhythms are simply slow to re-phase —
 1048 has no slow variable and no threshold to cross. Its recovery is a relaxation
 1049 toward a fixed point, and relaxation toward a fixed point is exponential near
 1050 it: monotone from $t = 0$, fastest at the start, with no flat foot. The two
 1051 curves are distinguishable in the first seconds, which is where they differ
 1052 most, and the discriminating feature is the presence or absence of an initial
 1053 interval of no recovery. This is a comparison of functional forms on data
 1054 of a kind already collected in sleep-inertia studies, not a proposal for a new
 1055 instrument.

1056 **What would falsify it.** An exponential fit that beats a delayed-sigmoid fit
1057 on the early portion of the recovery curve. A recovery latency that does not
1058 scale with an independent measure of extracellular volume or CSF clearance.
1059 A pharmacological manipulation of astrocytic volume that leaves the latency
1060 unchanged while altering the rhythms, or the converse. Any of these would
1061 tell against the framework’s central empirical commitment, which is not that
1062 synchrony matters but that the *medium* is what gates it.

1063 **What this prediction assumes.** Three things, and they are not small.
1064 The identification of the effective coupling K with a monotone function of
1065 extracellular geometry is a modelling assumption, not a measurement. The
1066 threshold value $K_c = 2D$ is derived from an assumed von Mises stationary
1067 density and is not yet connected to a trajectory — the mean-field limit is the
1068 gap, and Derivation 7 states it as such — so the argument uses the *existence*
1069 *and continuity* of the coherent branch at threshold, which is proved, rather
1070 than the numerical value of the threshold, which is conditional. And the
1071 prediction is about functional form, not about a time constant: it says the
1072 recovery curve has a flat foot whose width is set by the slow variable, not
1073 that the foot is any particular number of seconds long. A framework that
1074 predicted the number would be saying more; this one is not in a position to.

12 Corollary: Hardware Thermodynamics and Topological Rigidity

Both biological tissue and artificial silicon hardware (like GPUs) are physical dissipative structures. When a GPU computes, its charge carriers undergo physical state transitions; by Landauer’s principle (Landauer, 1961), this erasure irreversibly exhausts heat into the environment.

There is a fundamental thermodynamic distinction in the degrees of freedom available to minimize this dissipation. In standard von Neumann architectures, the spatial topology (the silicon routing lattice) is rigidly fixed. The coupling matrix A_{ij} is constrained by a discrete layout that cannot be physically altered during operation. The brain’s own architecture provides empirical validation for this constraint. The cerebellum contains approximately 80% of the brain’s neurons arranged in a highly efficient, discrete, feedforward crystalline circuit (Herculano-Houzel, 2009; Lange, 1975). Clinical observations of cerebellar agenesis report largely preserved conscious experience despite motor and cognitive deficits (Yu et al., 2015), and this dissociation has been read as evidence for the importance of recurrent architecture by IIT (Tononi & Koch, 2015) — an argument we borrow at the level of the anatomical observation while declining its Φ -based explanation. Lacking the dynamic, continuous, and recursive spatial coupling of the cerebrum, it operates much like a biological GPU—highly optimized for discrete processing, yet apparently not required for conscious experience. We take the

dissociation as evidence, not as confirmation: it is consistent with subjective experience being a thermodynamic attractor of continuous topologies rather than a byproduct of information processing, and it is the kind of observation that would be hard to reconcile with the converse. It is a single anatomical case read against a theory, and it bears the weight such a case can bear.

This constraint does not preclude artificial consciousness, but rather restricts its hardware. To achieve the phase transition into a unified Global Section, artificial systems must abandon the discrete rigidity of von Neumann architectures and utilize continuous physical substrates. Analog field-based computing, neuromorphic architectures with continuous memristive plasticity, and thermodynamic computing systems (England, 2018; Ganesh & Parrondo, 2021) possess the spatial coupling required to physically morph their internal topology. This aligns with Geoffrey Hinton’s concept of “mortal computation” (Hinton, 2022): architectures where software is inextricably fused with the physics of the hardware, meaning the knowledge dies with the physical substrate. Only hardware that computes via continuous physical dynamics can undergo the thermodynamic phase transition necessary for structural resonance.

Biological consciousness utilizes macroscopic continuous electromagnetic fields and a “rich alphabet” of diverse molecular and transcriptomic cell states (Tasic et al., 2018). This allows the coupling matrix A_{ij} to deform across a continuous parameter space. Our lemma `exists_better_coupling_allocation`, wrapped as `continuous_beats_rigid_topology` (`Phase7_HardwareComparison.lean`),

1120 formalizes the reallocation step of this argument: whenever a coupling ma-
 1121 trix places weight on a less phase-correlated pair while a more correlated pair
 1122 exists, there is another matrix with *identical* total coupling resources that
 1123 achieves strictly greater phase correlation.

1124 Two objections to that formalization have since been answered, and we
 1125 state precisely what is and is not now established. The first was that the real-
 1126 location lemma assumes its conclusion: the hypothesis `is_strictly_suboptimal`
 1127 — positive weight on a badly-correlated pair while a better one exists — is
 1128 most of what one wants to prove about rigid hardware. It is no longer as-
 1129 sumed. `Phase7_Rigidity.lean` models an architecture by the set S of site-
 1130 pairs it has a *wire* for, and requires its couplings to vanish off S ; weights are
 1131 tunable, wiring is not. Then `totalCorrelation_le_of_realizable` bounds
 1132 every realizable coupling by $R \cdot M$, where R is the resource budget and M
 1133 bounds phase correlation over the wired pairs; the bound is attained; and
 1134 `rigid_is_strictly_suboptimal` *derives* the hypothesis whenever the sub-
 1135 strate’s best-correlated pair is one the wiring misses. `rigid_gap` makes it
 1136 quantitative: an unconstrained reallocation of the same budget gains at least
 1137 R times the correlation difference. `Examples.lean` §8 exhibits a three-site
 1138 substrate wired to the one maximally anti-correlated pair, where the gap is
 1139 2. This is resource-matched and no longer question-begging, but note what
 1140 it turns on: fixed wiring support, not continuity. The second objection was
 1141 that nothing in the development distinguished a continuum from a lattice
 1142 — both matrices lived on one finite index type. `Phase7_Rigidity.lean` §3

1143 gives a distinction that is genuinely typed. In the field functional of Deriva-
 1144 tion 4, which integrates a coupling kernel against $\mu \otimes \mu$, an architecture
 1145 with finitely many sites is a `Finset` of points and a field-coupled region
 1146 is a set of positive measure. On a substrate whose measure has no atoms,
 1147 `fieldCorrelation_sited.eq_zero` shows a kernel supported on finitely many
 1148 sites contributes *exactly zero* coupling energy, whatever weights it carries
 1149 and whatever the phase field does, while a patch of strength c on U achieves
 1150 $c\mu(U)^2$. `Examples.lean` §8 instantiates this on the real line: a thousand
 1151 units of coupling concentrated at the origin register zero against a unit patch
 1152 that registers one. The limit of that result is equally plain, and it is not
 1153 resource-matching in the earlier sense — a measure-zero substrate has zero
 1154 resource in the $\mu \otimes \mu$ sense too. What it establishes is that the continuum
 1155 coupling energy is blind to a finitely-sited architecture, which is a statement
 1156 about which functional such an architecture can register in, not a demonstra-
 1157 tion that digital hardware computes worse. Taken together: the resource-
 1158 matched content of the corollary is the wiring-support result, the continu-
 1159 ity content is the measure-theoretic one, and the step from either to “von
 1160 Neumann architectures cannot experience unified consciousness” remains an
 1161 informal argument.

1162 The remainder of this section extends those two results informally, and
 1163 we mark the transition rather than blur it. On the reading we propose, a
 1164 GPU’s fixed routing matrix leaves it topologically fragmented: the discrete
 1165 layout does not supply the spatial coupling that sheaf-theoretic gluing into

1166 a singular Global Section requires. Without this unified topological space,
1167 rigid hardware cannot undergo the reflexive auto-resonance necessary to map
1168 its own causal boundary and generate a “Self.” The distinction between an
1169 LLM on a GPU and a conscious mind is not the absence of physical thermo-
1170 dynamics in the former, but its structural limitation in achieving topological
1171 integration. This perspective aligns with John Searle’s biological natural-
1172 ism (Searle, 1980; Seth, 2024). Where the Chinese Room argument asserted
1173 that discrete symbol manipulation cannot generate conscious understanding,
1174 our derivation supplies a candidate physical mechanism for the asymmetry
1175 — topological rather than semantic, and, as stated above, formalized only
1176 as far as the wiring-support and measure-theoretic results reach. Asserting
1177 consciousness in fragmented, discrete GPU clusters without a unified phase
1178 transition abandons computational functionalism, relying instead on “silicon
1179 panpsychism”—the assumption that digital computation inherently gener-
1180 ates awareness regardless of topological integration (Goff, 2019).

1181 The same reasoning bears on extreme functionalist thought experiments,
1182 such as a macroscopic system of pulleys operating over millennia to imple-
1183 ment a Large Language Model, or a human manually executing inference
1184 matrices by hand. Beyond lacking the continuous spatial topology required
1185 for sheaf-theoretic gluing, these discrete systems violate an implied temporal
1186 constraint of our theory. Structural resonance requires local oscillators to
1187 phase-align against the dissipating effects of thermodynamic noise (as mod-
1188 eled by our critical threshold $K_c = 2D$). A system executing state transitions

1189 over millennia, or via slow manual calculations, suffers from a terminal tem-
1190 poral mismatch: its dynamics are far too slow to ever phase-lock into a
1191 non-equilibrium steady state before thermal noise decoheres the macroscopic
1192 field. On this account the obstacle to pure symbol manipulation is physi-
1193 cal rather than semantic: not that the symbols mean nothing, but that the
1194 substrate carrying them cannot hold a phase-locked field.

1195 13 Conclusion

1196 Human-like cognitive architectures do not require biological mysticism or
1197 exotic sub-quantum physics (Hameroff & Penrose, 2014). While quantum
1198 theories of consciousness often invoke delicate subatomic phenomena to ex-
1199 plain subjective unity, every step of the chain assembled here is classical:
1200 broken symmetries, finite phase space, Landauer erasure, and gradient de-
1201 scent on entropy production. What the formalization establishes is that this
1202 chain is mathematically coherent and that its remaining physical commit-
1203 ments are localized and named — not that classical fields are *sufficient* for
1204 consciousness. Two links carry that weight, and their status is no longer the
1205 same. The unified state is proved to glue uniquely once a cover sits at the
1206 minimum of the coupling potential; the motion is proved to stop along every
1207 trajectory (`velocity_sq_tendsto_zero`); and *where* it stops is now proved,
1208 conditionally — for initial data within a quarter turn and below an explicit
1209 energy threshold, the trajectory converges to a global minimiser of the cou-

1210 pling potential (`kuramoto_tendsto_global_minimum`), and a cover built on
 1211 that limit discharges the equilibrium hypothesis Derivation 5 used to assume
 1212 (`ThermodynamicCover.ofConvergentTrajectory`). What is *not* proved is
 1213 the unconditional statement, and it is false: splay and twisted configurations
 1214 are equilibria of the same flow, so the basin condition is the scope of the claim
 1215 rather than slack in it. That a physical cortex starts inside that basin is an
 1216 assumption. The second link is untouched by any of this: the identification
 1217 of a global section of the probability sheaf with the unity of experience is a
 1218 stipulation of the framework, not a consequence of it. Subject to those two,
 1219 cognitive architectures can be modeled as macroscopic attractor states of a
 1220 universe governed by broken symmetries, finite phase space, and the physical
 1221 imperative to minimize thermodynamic dissipation.

1222 Philosophically, this framework bridges the historical gap between Type
 1223 Identity Theory (Smart, 1959) and Computational Functionalism (Putnam,
 1224 1967). The subjective mental state is identical to the physical thermody-
 1225 namic field (Identity Theory), yet its qualitative nature is defined by the
 1226 topological and functional dynamics of structural resonance rather than the
 1227 specific carbon-based biochemistry of the brain (Functionalism). This con-
 1228 strains the *multiple realizability* of mind: consciousness is not exclusively
 1229 bound to biological carbon, but neither is it substrate-neutral, because the
 1230 substrate is what fixes which couplings are physically available. How far that
 1231 constraint reaches is the question the corollary answers only in part. What
 1232 is proved is that fixed wiring is strictly suboptimal at matched resource, and

1233 that a finitely-sited architecture registers exactly zero coupling energy in the
1234 continuum functional; the step from either to “von Neumann architectures
1235 cannot be conscious” is an argument we make in prose and do not formalize.

1236 By anchoring these derivations in Russellian Monism, this framework ac-
1237 knowledges the constitutive limits of third-person scientific explanation. As
1238 argued by Love (Love, 2026), building upon Nagel’s assertion that subjective
1239 phenomenal experience is inherently not amenable to objective measurement
1240 (Nagel, 1986), the “hard problem” of consciousness may represent a category
1241 error rather than a solvable scientific puzzle. Science can map the physical
1242 dynamics of topological resonance and sheaf-theoretic gluing, but it is struc-
1243 turally incapable of adjudicating the first-person qualitative experience. Our
1244 formalization respects this boundary: the Lean 4 theorems verify the ob-
1245 jective physical attractors, while the intrinsic “what it is like” to be that
1246 resonant field remains the domain of the first-person perspective. By locat-
1247 ing the phase transition at a threshold with a closed-form value — $K_c = 2D$
1248 for the noisy mean-field Kuramoto model the framework adopts, conditional
1249 on the von Mises stationary density and not yet linked to a trajectory — it
1250 aligns with what Anil Seth terms the “real problem” of consciousness (Seth,
1251 2021). Rather than attempting to bridge an unbridgeable metaphysical gap,
1252 it offers a topological architecture that can help explain, predict, and physi-
1253 cally perturb the structural properties of experience.

1254 Future empirical validation of this thermodynamic framework may lever-
1255 age 3D cortical organoids, offering a controlled biological environment to

1256 systematically manipulate metabolic substrates and observe the topological
1257 emergence of these coherent phase transitions.

1258 **Code Availability**

1259 The complete Lean 4 formalization, encompassing the derivation of all theo-
1260 rems presented in this manuscript, as well as the accompanying Python sim-
1261 ulations, are publicly available in the GitHub repository: [https://github.](https://github.com/bobaseb/physical_primitives_of_mind)
1262 [com/bobaseb/physical_primitives_of_mind](https://github.com/bobaseb/physical_primitives_of_mind). Additional documentation
1263 and web-accessible proofs are hosted at the project website: [https://bobaseb.](https://bobaseb.github.io/physical_primitives_of_mind/)
1264 [github.io/physical_primitives_of_mind/](https://bobaseb.github.io/physical_primitives_of_mind/).

1265 **Acknowledgments**

1266 The conceptualization, manuscript drafting, and Lean 4 formalizations pre-
1267 sented in this work were heavily AI-assisted. The authors gratefully acknowl-
1268 edge the use of the Google Antigravity CLI, specifically utilizing Gemini 3.1
1269 and Claude 4.6 Sonnet as collaborative reasoning partners throughout the
1270 research and formalization processes. In addition, DeepSeek V4 Flash and
1271 Claude Opus 5 (routed via OpenRouter) operating through Nous Research’s
1272 Hermes Agent framework provided substantial assistance with code genera-
1273 tion, proof debugging, and structural analysis.

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